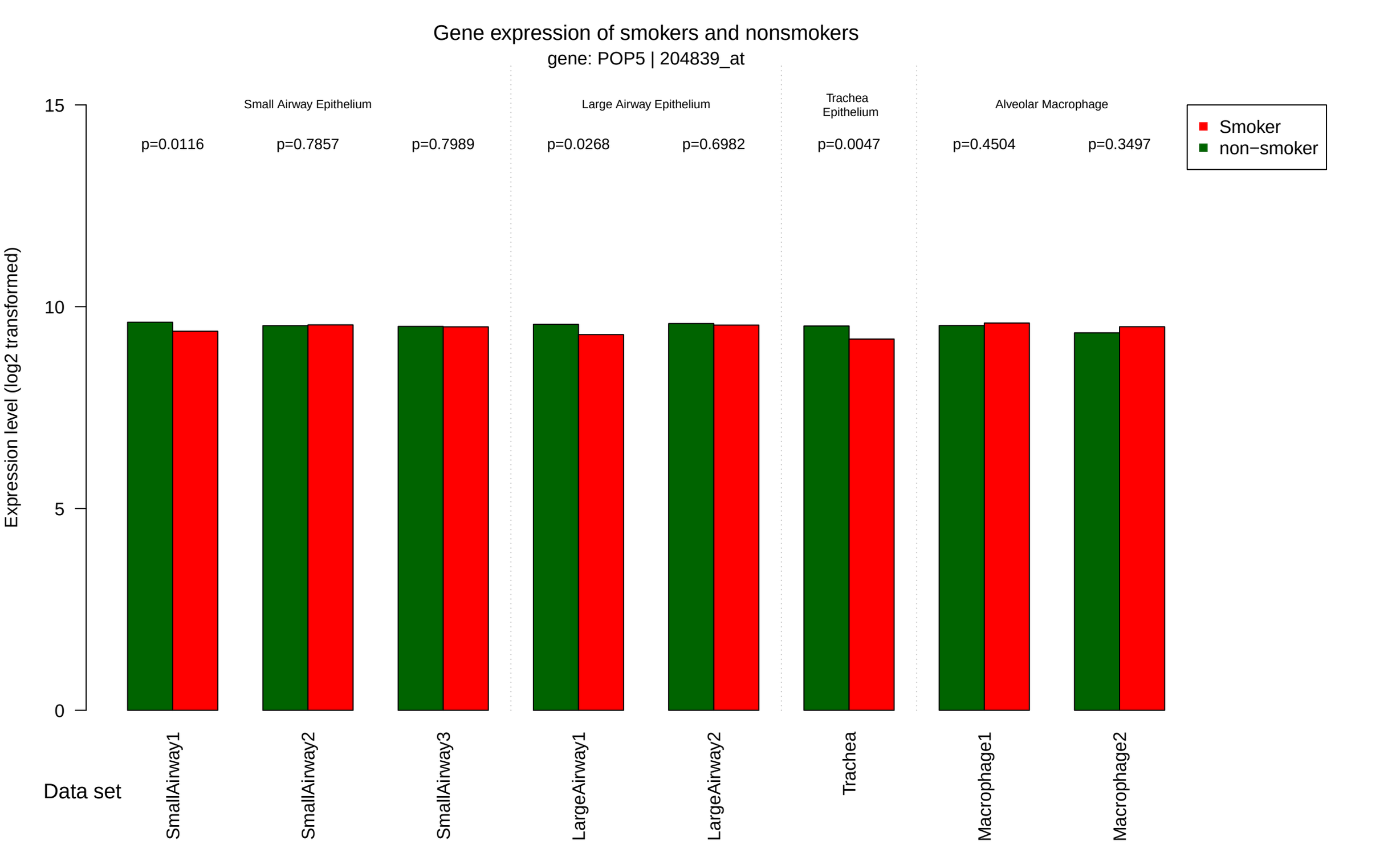
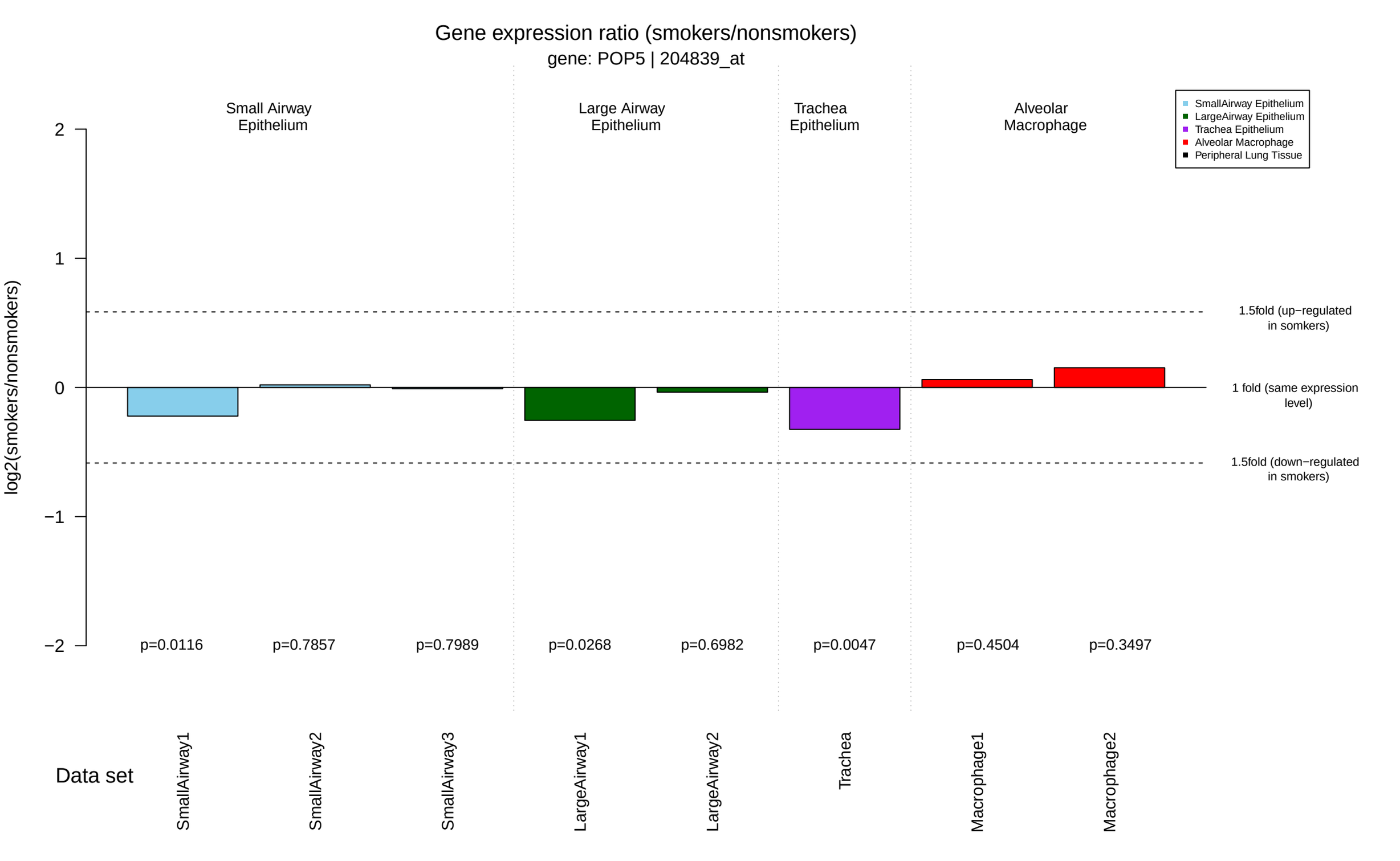
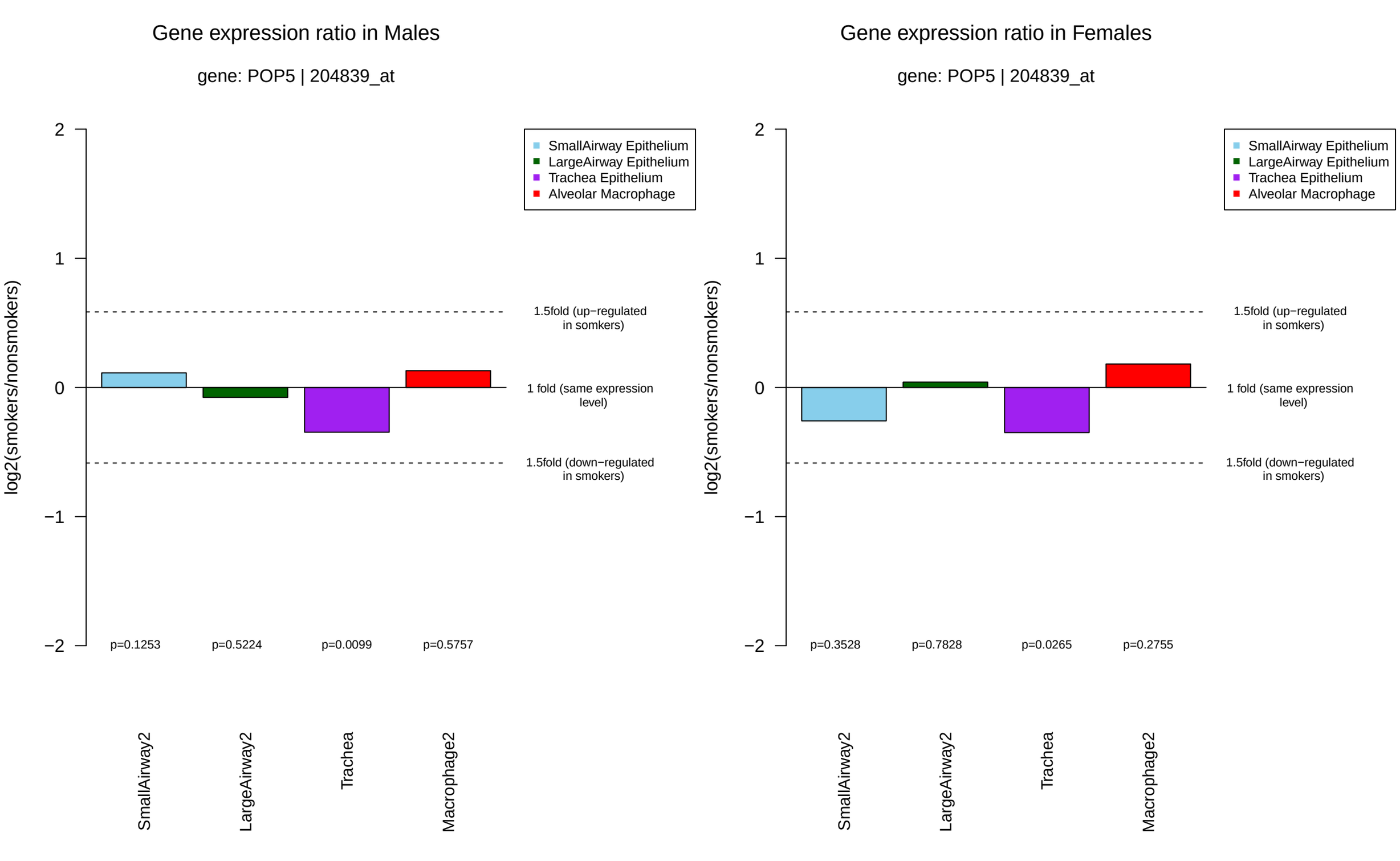
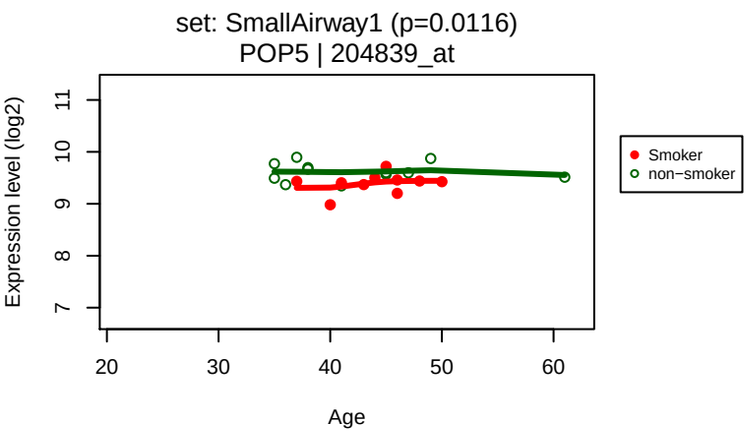
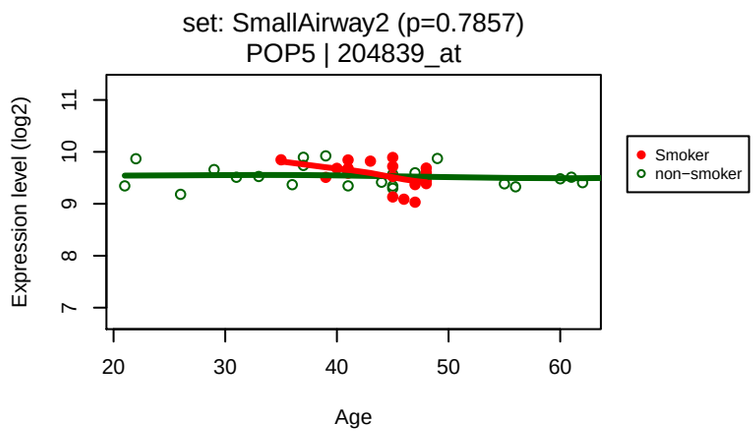
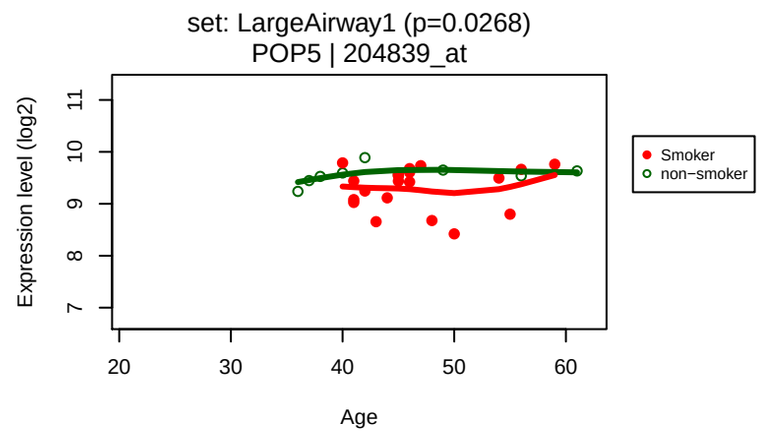

Gene expression of smokers and nonsmokers
gene: POP5 | 204839_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0116
p=0.7857
p=0.7989
p=0.0268
p=0.6982
p=0.0047
p=0.4504
p=0.3497
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

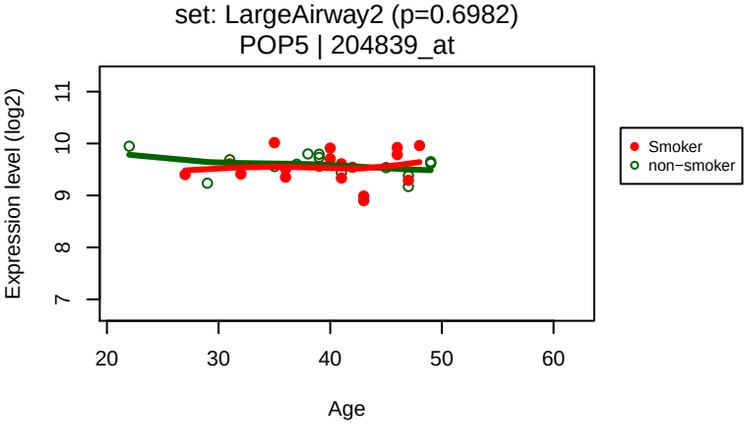

Gene expression ratio (smokers/nonsmokers)
gene: POP5 | 204839_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0116
p=0.7857
p=0.7989
p=0.0268
p=0.6982
p=0.0047
p=0.4504
p=0.3497
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

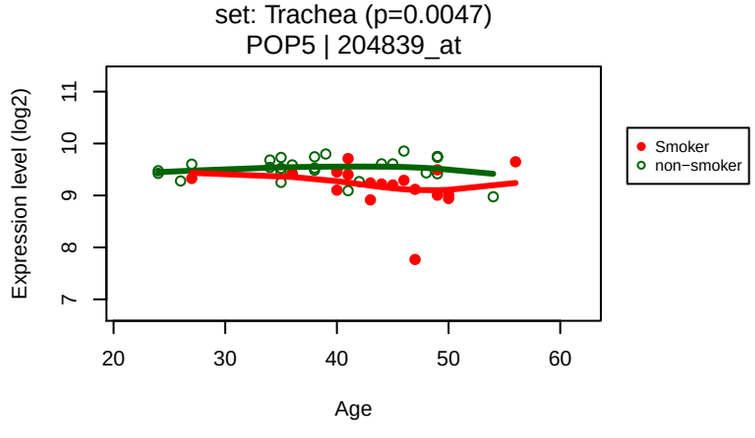

Gene expression ratio in Males
gene: POP5 | 204839_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1253
p=0.5224
p=0.0099
p=0.5757
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POP5 | 204839_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3528
p=0.7828
p=0.0265
p=0.2755
SmallAirway2
LargeAirway2
Trachea
Macrophage2

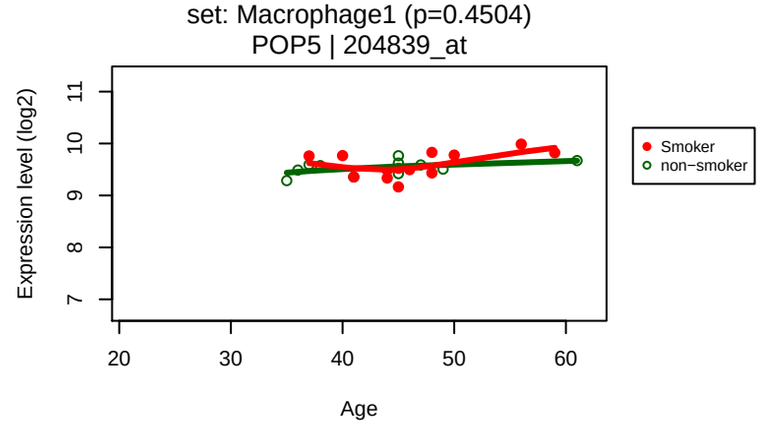

set: SmallAirway1 (p=0.0116)
POP5 | 204839_at
Expression level (log2)
Age
Smoker
non−smoker

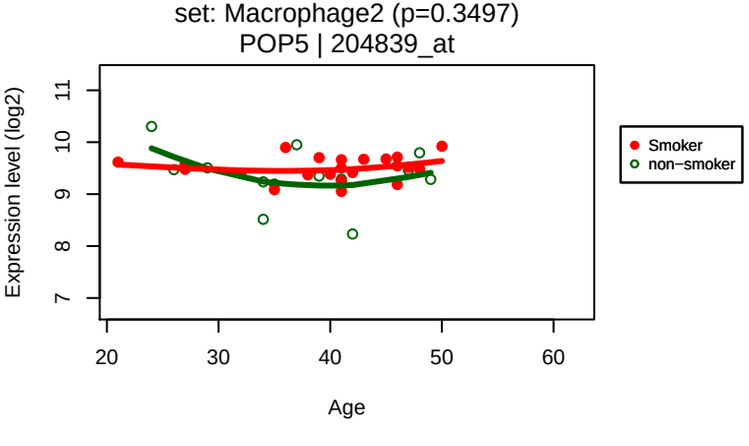

set: SmallAirway2 (p=0.7857)
POP5 | 204839_at
Expression level (log2)
Age
Smoker
non−smoker

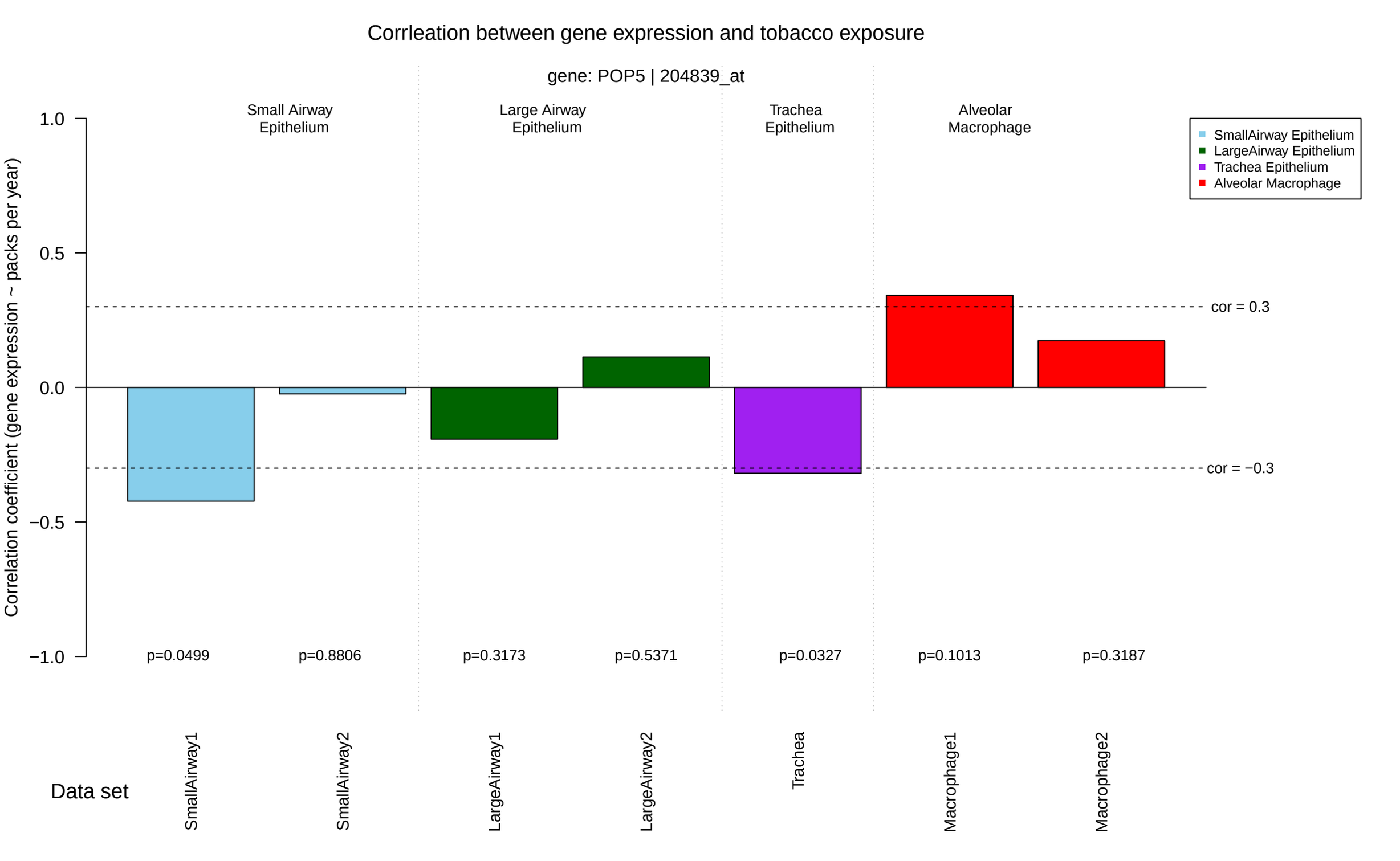

set: LargeAirway1 (p=0.0268)
POP5 | 204839_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6982)
POP5 | 204839_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0047)
POP5 | 204839_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4504)
POP5 | 204839_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3497)
POP5 | 204839_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POP5 | 204839_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0499
p=0.8806
p=0.3173
p=0.5371
p=0.0327
p=0.1013
p=0.3187
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2